

MedVerify AI — Final Complete Implementation Plan & Architecture

Single Unified Build Sequence — Reliability-Aware, Population-Aware, Faithfulness-Guarded Medical Claim Verification

Consolidates the base architecture, the R1–R4 remediation findings, and the population-applicability research contribution into one continuous plan.

Core Hypothesis and Contribution

Health-misinformation verification improves when retrieved evidence is ranked not only by semantic relevance, but by source reliability (R_i) and population applicability (P_i) jointly, combined as $W_i = R_i \times P_i$, and when the resulting explanation is independently verified for faithfulness before being shown to a user. No single component of this pipeline is individually novel in the current literature (2024–2026); the tested contribution is the specific combination, its calibration, and its measured effect on verification accuracy — evaluated against conventional and reliability-only baselines.

1. Outstanding Validation Gate (read this before anything else)

The project has repeatedly built ahead of validating what it built. This table is the current honest status of every open item across the whole project. Nothing below is closed until the stated evidence exists — a working feature is not the same as a validated one.

Item	What's Built	What's Still Missing	Status
R1 — Dataset composition	PubHealth ingestion script implemented	Post-ingestion class balance across Diabetes/Cardiovascular/Vaccination not reported	OPEN — report the numbers
R2 — Classifier validity	Model trained (originally BioBERT, now reportedly DistilBERT)	Duplicate/leakage check between train/test not re-run; per-class F1 on corrected data not reported; model choice (BioBERT vs DistilBERT) not justified	OPEN — re-audit and confirm model choice
R3 — Retrieval quality	Retrieval pipeline live (now on MiniLM embeddings, not BGE-large)	nDCG@10 vs. SciFact never formally computed on the current embedding model	OPEN — run the real evaluation
R4 — Faithfulness model licensing	Switched to public cross-encoder/nli-deberta-v3-small	Confirm this was the deliberate fix; general-domain NLI may underperform a medical-tuned model — measure, don't assume	LIKELY RESOLVED — confirm and measure
P1–P4 — Population-applicability validation	P_i scoring implemented and shown in UI	Fraction of claims with real population data, extraction accuracy audit, human-labeled ground truth (κ check), and age-format handling all previously agreed but not reported	OPEN — all four still required
Calibration (ECE / Brier)	Not yet implemented	This is the project's strongest claimed differentiator vs. VERIRAG/MedRAGChecker and has never been computed	OPEN — required before any novelty claim is finalized
Medical-safety refusal routing	Built earlier (Stage 12 equivalent)	Not mentioned in the most recent build report — confirm it still exists and still passes its test set	CONFIRM NOT LOST
Two-pass review (Engineering + Research Validity)	Never formally convened	Cannot be convened until every row above is closed	BLOCKED

2. Complete System Architecture

One continuous pipeline. The earlier “medical courtroom” framing is kept because it maps cleanly onto the real stages: Triage (classification) → Investigation (retrieval) → Demographic Analysis (population applicability) → Jury Deliberation (reliability + population-weighted consensus) → Fraud Inspection (faithfulness guard) → Verdict.



3. Unified Implementation Plan (single sequence, no phase split)

Fifteen stages, one continuous build order. Every stage that already has working code is marked so in its Exit Criteria as “re-verify,” not “build from scratch” — this plan assumes the current codebase as a starting point, not a blank slate.

Stage 0 — Pre-Build Validation (Reliability + Population + Faithfulness) — Week 1

Objective: Confirm, on paper, that the scoring formula and faithfulness detection approach work before trusting them in production.

Key Tasks

- Reliability score vs. SciFact ground truth: check correlation direction on 30–50 examples.
- Faithfulness (NLI + hedge) vs. 20–30 manually labeled LLM explanations.
- Population coverage (P1): what fraction of real claims specify age/sex/region.
- Population extraction accuracy (P2) and ground-truth applicability labeling with inter-rater kappa (P3).
- Age-format survey and scoring rules for range / mean±SD / qualitative terms (P4).

Exit Criteria

- All five sub-checks produce a documented number, not an assumption; any check showing no signal triggers a redesign before Stage 1.

Stage 1 — Architecture & Contracts — Week 1–2

Objective: Lock API contracts, state machine, and version-metadata fields (model, embedding, KB, ranking, prompt, faithfulness-model versions) across the whole pipeline.

Key Tasks

- Define /api/claims and /api/verifications contracts.
- Define the CREATED → ... → COMPLETED/FAILED state machine.
- Define the version-metadata object as mandatory on every verification record.

Exit Criteria

- Contract frozen and reviewed before further implementation.

Stage 2 — Backend Foundation — re-verify — Week 2

Objective: Confirm the existing FastAPI + PostgreSQL/SQLite + Docker Compose + Nginx foundation meets health-check and request-tracing requirements.

Key Tasks

- Confirm /health endpoints on all services.
- Confirm request-ID propagation through Nginx → FastAPI → downstream calls.
- Confirm JWT + PBKDF2-HMAC-SHA256 auth is applied to all non-public endpoints.

Exit Criteria

- docker compose up brings all services healthy; a request ID is traceable end-to-end.

Stage 3 — Stable API / Frontend Integration — re-verify — Week 2–3

Objective: Confirm the React 18 + Vite + Tailwind + Zustand frontend, including the 5-gauge dashboard and SSE progress stream, works against both mock and live data paths.

Key Tasks

- Confirm SSE events stream at the intended cadence and every state-machine stage renders.
- Confirm the FAILED state renders a clear, non-technical error.

Exit Criteria

- Full submit-to-result flow works end-to-end on both mock and live backends with no contract mismatch.

Stage 4 — Dataset Governance — close R1 — Week 3

Objective: Finalize dataset composition using the now-implemented PubHealth ingestion, correcting the original CoAID-only imbalance.

Key Tasks

- Report post-ingestion class counts for Diabetes, Cardiovascular Disease, and Vaccination.
- Confirm CoAID is scoped only to the Vaccination category, not treated as general health data.
- Run the medical-consultant label audit (50–100 samples) on the newly ingested PubHealth data specifically.
- Hash and freeze train/validation/test manifests.

Exit Criteria

- Each disease category has a reasonably balanced sample size (no category under-represented by an order of magnitude); label-audit disagreement rate documented.

Stage 5 — Knowledge Base Construction — re-verify embedding choice — Week 3–4

Objective: Confirm the knowledge base (WHO/CDC curated + indexed PubMed/PMC) is built cleanly on whichever embedding model is finalized, with no leakage against Stage 4's frozen test sets.

Key Tasks

- Decide and document: MiniLM (already implemented, faster/lighter) vs. BGE-large (original recommendation, stronger biomedical retrieval) — or keep MiniLM as the served default with BGE-large benchmarked as an upgrade candidate in Stage 7.
- Run the leakage check between indexed evidence and frozen test manifests.

Exit Criteria

- Knowledge base queryable end-to-end; leakage check documented as zero overlap.

Stage 6 — Claim Extraction, Disease Classification, Population Fields — Week 4–5

Objective: Extract the structured claim (disease, intervention, outcome) and, in the same pass, extract population fields with confidence scores, defaulting to null when unspecified.

Key Tasks

- Confirm/replace the classification model: re-audit whether DistilBERT or BioBERT is in production, and re-run evaluation on whichever is chosen.
- Apply the Stage 0 (P4) age-format rules uniformly to claim-side population extraction.

Exit Criteria

- Macro-F1 ≥ 0.85 on the R1-corrected, leakage-checked test split, per-class recall reported individually; population-field extraction rate matches the Stage 0 (P1) measured percentage.

Stage 7 — Retrieval + Study Metadata Extraction — Week 5–6

Objective: Retrieve candidate evidence (FAISS + live PubMed/PMC) and extract structured study metadata, including population fields, for every candidate.

Key Tasks

- Apply query normalization, deduplication, filtering, caching, and PubMed timeout/retry/fallback behavior.
- Apply the Stage 0 (P2) extraction-accuracy findings: any population field below the accuracy threshold is treated as unspecified rather than trusted.

Exit Criteria

- nDCG@10 ≥ 0.75 against SciFact, measured on the actual production embedding model — not a 3-query smoke test.

Stage 8 — Reliability + Population-Weighted Evidence Ranking — Week 6–7

Objective: Compute R_i (source reliability) and P_i (population applicability) independently, then combine as $W_i = R_i \times P_i$ for final ranking.

Key Tasks

- Implement R_i as a documented, versioned function (source tier, recency decay, evidence type).
- Implement P_i using the Stage 0 (P3/P4) validated rules — age/sex/region compatibility, never defaulting mismatched evidence to zero.
- Compare $W_i = R_i \times P_i$ against an additive alternative ($R_i + P_i$, normalized) as a documented design choice, not an assumption.

Exit Criteria

- Combined ranking evaluated against the Stage 0 (P3) human-labeled applicability set; multiplicative vs. additive combination compared with a stated result, not just implemented.

Stage 9 — Consensus Engine (Jury Deliberation) — Week 7

Objective: Represent supporting/contradicting/neutral evidence using W_i -weighted consensus, shown separately from the raw evidence-count distribution.

Key Tasks

- Compute $\text{weighted_consensus} = \Sigma(W_i \times \text{stance}_i) / \Sigma(W_i)$.
- Expose both `evidence_distribution` (raw) and `weighted_consensus` (W_i -weighted) as separate, clearly labeled API fields.

Exit Criteria

- Manual review of 20–30 cases confirms weighted consensus diverges correctly from raw count when reliability or population applicability clearly diverge from quantity.

Stage 10 — Explanation Generation — Week 7–8

Objective: Generate a cited, certainty-preserving explanation grounded only in the ranked, weighted evidence.

Key Tasks

- Require per-sentence evidence citation in the generation prompt.
- Require the prompt to preserve the evidence's original certainty language (no upgrading “associated with” to “causes”).

Exit Criteria

- Sample review confirms full citation coverage and no certainty inflation in generated text.

Stage 11 — Faithfulness Verification (Fraud Inspection) — confirm R4 fix — Week 8

Objective: Verify every explanation sentence against its cited evidence using NLI plus the BioScope hedge/certainty guard.

Key Tasks

- Confirm cross-encoder/nli-deberta-v3-small is the deliberate, documented replacement for the MedNLI-licensing-blocked approach.
- Run the combined NLI + hedge detector against the 200-example manually annotated hallucination set.
- Report NLI-only recall vs. NLI+hedge recall separately to show the hedge layer's added value.

Exit Criteria

- Faithfulness recall ≥ 0.90 on the annotated set; flagged sentences are regenerated or marked “unverified,” never silently shown as fact.

Stage 12 — Credibility Indicator, Calibration, and Medical Safety — Week 8–9

Objective: Produce a componentized, calibrated Credibility Indicator (feeding the 5-gauge UI) and confirm medical-safety refusal routing is intact.

Key Tasks

- Compute ECE and Brier score for the credibility output against ground-truth verdicts — not yet done anywhere in the project and currently the single most important missing evaluation.
- Re-run the diagnosis/treatment/emergency-advice-seeking safety-routing test set (15–20 prompts) to confirm it was not lost in the recent refactor.
- Ensure UI language reads “System Confidence Indicator,” never “probability the claim is true.”

Exit Criteria

- Calibration curve and ECE/Brier reported; 100% correct safety-routing on the test set reconfirmed.

Stage 13 — Reliability + Security Hardening — re-verify — Week 9

Objective: Confirm rate limiting, prompt-injection resistance, and graceful degradation hold under the current Docker/Nginx deployment.

Key Tasks

- Re-run the 10–15 prompt-injection test cases against the current build.
- Simulate a PubMed outage and confirm graceful fallback, not a hang or crash.
- Confirm JWT/PBKDF2 auth covers every non-public endpoint, including any added since the last security pass.

Exit Criteria

- Zero verdict change under prompt-injection tests; documented graceful degradation under simulated outage.

Stage 14 — Full Research Evaluation + Two-Pass Review — Week 9–11

Objective: Run the complete baseline comparison and convene both review gates — the step the project has consistently deferred.

Key Tasks

- B0: LLM only, no retrieval.
- B1: Conventional relevance-based RAG.
- B2: Reliability-aware RAG (R_i only, no population term) — the VERIRAG-comparable baseline.
- B3 (proposed): Full system, $W_i = R_i \times P_i$, with faithfulness verification.
- Compute verdict-change rate between B2 and B3 specifically — this isolates the population term's real contribution.
- Recruit medical-consultant review of 100–200 claims for expert-agreement rate.

Exit Criteria

- B3 outperforms B0–B2 on accuracy/macro-F1 with the gap attributable to specific components via ablation; $\geq 90\%$ expert agreement; both Engineering Reliability and Research Validity reviews pass in full before this project is called complete.

4. Datasets (Final)

Dataset	Role	Status
PubHealth	Primary classification fine-tuning + evaluation; now the fix for Diabetes/Cardiovascular coverage	Ingestion script implemented — post-ingestion class balance still needs to be reported (Stage 4)
SciFact	Evidence-ranking (R_i) and retrieval (nDCG@10) calibration	Available; formal nDCG@10 evaluation still not run (Stage 7)
CoAID	Vaccination category only — confirmed COVID/vaccine-scoped, not general health data	In use; scope now correctly documented
MedNLI	Originally planned for faithfulness fine-tuning	Superseded by public cross-encoder/nli-deberta-v3-small — licensing risk avoided, confirm this is deliberate
Stage 0 / P3 population-applicability set	Ground truth for population-match accuracy	Methodology defined; labeling not yet executed

5. Models (Final, With Open Decisions Flagged)

Component	Currently Implemented	Open Decision
Disease Classification	DistilBERT (per latest build report)	Confirm vs. BioBERT; re-evaluate macro-F1 on R1-corrected data before finalizing either choice
Retrieval Embedding	all-MiniLM-L6-v2 (384-dim)	Benchmark against BAAI/bge-large-en-v1.5 on the actual nDCG@10 evaluation (Stage 7) before accepting MiniLM as final
Faithfulness NLI	cross-encoder/nli-deberta-v3-small (public)	Confirmed license-safe; measure whether accuracy is sufficient vs. a medical-tuned alternative
Hedge/Certainty Detection	BioScope-corpus-based regex guard	Working as designed; no open decision
Explanation LLM	Open-weight (Qwen 3 / Llama 3.1 / Gemma 3 family, per original plan)	Confirm which specific model is in production and its version is logged per verification
Population Extraction	NLP extraction pipeline for age/sex/region	Accuracy not yet audited (Stage 0, P2) — required before trusting P _i in evaluation

6. Evaluation Plan (Final)

Metric	Target	Status
Disease classification macro-F1	≥ 0.85	Not yet re-measured on corrected data + confirmed model
Retrieval nDCG@10 (vs. SciFact)	≥ 0.75	Not yet formally run on production embedding model
Faithfulness recall (NLI+hedge)	≥ 0.90	Not yet run on the current cross-encoder model
Calibration (ECE / Brier)	Reported and reviewed, no fixed target yet	Not yet implemented anywhere in the project
Population-match accuracy	Reported against P3 human-labeled set	Ground-truth set not yet built
Expert agreement (end-to-end)	≥ 90%	Not yet run
B2 vs. B3 verdict-change rate	Reported descriptively	Requires B0–B3 baseline suite, not yet run

7. Technology Stack (Final, as Actually Built)

Layer	Technology
AI/ML	DistilBERT (classification, pending re-confirmation), all-MiniLM-L6-v2 (embeddings, pending benchmark), cross-encoder/nli-deberta-v3-small (faithfulness), FAISS, BioScope-based hedge detection
Backend	Python 3.12, FastAPI, Uvicorn, SQLAlchemy 2.0, PostgreSQL/SQLite
Frontend	React 18, Vite, Tailwind CSS, Zustand, Recharts, SSE for live progress
DevOps	Docker, Docker Compose, Nginx reverse proxy
Security	PBKDF2-HMAC-SHA256, JWT token auth, rate limiting, prompt-injection sanitization

8. Summary

The architecture is complete and the engineering is largely production-shaped: containerized, authenticated, streaming live progress, and now demographically aware. What remains is not more building — it is closing Section 1's validation gate. Every number in Section 6 is currently either missing or unverified on the current codebase. The single next action, before any further

feature work, is Stage 0: run the five pre-build validation checks against the system exactly as it exists today, then proceed stage by stage through Stage 14 until the two-pass review actually convenes and passes. A demographically-aware, faithfulness-guarded verification system that has never been calibrated is a strong prototype; closing this gate is what turns it into a defensible research contribution.